

Emotional distress predicts change in adolescent problematic digital use and related symptoms

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Running title: Temporal ordering of adolescent digital use

Abstract

Whether adolescent problematic digital use precedes or follows broader mental health difficulties remains unresolved. In the School-based Evaluation Advancing Response for Child Health (SEARCH) cohort in China (N = 10,356; n = 6,092 reassessed one year later), a cross-lagged panel network (CLPN) showed that emotional distress had the strongest positive net directionality (+0.249, 95% CI 0.086 to 0.418) and predicted later problematic digital use, insomnia, broad difficulties and diet. Problematic digital use showed no significant net directionality. In the UK Millennium Cohort Study (n = 7,838), social media use was the strongest net sink (−0.211, 95% CI −0.260 to −0.139), whereas hyperactivity/inattention was the leading net source. Static structural-control positions differed from CLPN net directionality. Bayesian net out-degree showed a positive but imprecise association with CLPN net directionality. In the Global Mind Project (n=147,506), emotion had the highest cross-sectional network strength. Across longitudinal cohorts, psychological difficulties occupied leading net source positions, whereas screen-related measures did not.

Keywords: adolescent mental health; emotional distress; problematic digital use; social media use; cross-lagged panel network; network control

Introduction

Mental health difficulties are widespread among young people¹. Within this context, adolescent problematic digital use rarely occurs in isolation. Problematic internet or smartphone use is associated with emotional distress and broader behavioural difficulties²⁻⁴, while greater social-media and screen exposure has been linked particularly to depressive symptoms and sleep disturbance⁵⁻⁹. These associations are clinically important, but they do not establish temporal order. Emotional distress may precede compensatory or avoidant digital engagement, whereas escalating digital use may displace sleep, physical activity or offline interaction. Longitudinal estimates are often small and vary across digital-use constructs,

developmental periods and analytical specifications^{10, 11}. The central question is which domains predict subsequent change across the wider mental health system.

Network models provide a framework for examining this ordering without reducing multidomain comorbidity to a single total score¹²⁻¹⁵. Rather than treating co-occurring difficulties as manifestations of a single underlying disorder, the network approach conceptualises symptoms and health domains as connected components of a wider system^{16, 17}. Cross-sectional Gaussian graphical models estimate conditional associations, while centrality measures identify highly connected nodes^{18, 19}. Connectedness alone does not indicate structural leverage or temporal priority. Structural-control methods address the static architecture of the system: minimum dominating set (MDS) analysis identifies the smallest sets of nodes whose neighbourhoods cover the estimated network, whereas module-control analysis characterises inter-module structural coverage²⁰⁻²². These topological quantities may reveal structural leverage, but they do not establish causal effects or temporal precedence²³⁻²⁵. Cross-lagged panel networks (CLPNs) instead quantify prospective prediction among domains, allowing static structural position to be compared with CLPN net directionality²⁶.

Establishing which domains lead change over time has direct implications for how adolescent problematic digital use is understood and addressed. If problematic digital use predicts subsequent broader difficulties, early assessment and prevention should address digital behaviour directly. If emotional distress predicts later problematic digital use, problematic digital use should be understood within a broader pattern of emotional difficulties.

Against this background, we used a CLPN in the School-based Evaluation Advancing Response for Child Health (SEARCH) cohort in Jiangsu, China²⁷ to identify the leading longitudinal predictors across six health domains, with particular focus on the relative positions of problematic digital use and emotional distress. The model included generalised problematic digital use, emotional distress, broad Strengths and Difficulties Questionnaire (SDQ) difficulties, diet, insomnia and physical activity in 6,092 adolescents reassessed approximately one year later. We then examined whether the longitudinal position of screen-related measures extended to the UK Millennium Cohort Study (MCS), where social media use and gaming were measured directly between ages 14 and 17²⁸. Next, we characterised the 104-item baseline SEARCH network in 10,356 adolescents and compared its structural-control nodes with the CLPN net source positions. Within SEARCH, an exploratory Bayesian graph was used to assess whether baseline net out-degree aligned with the longitudinal ordering observed in the CLPN. Finally, a pooled multinational young-adult sample from the Global Mind Project (GMP)^{29, 30} provided a cross-sectional comparison of network strength of emotion and social media use. Across the longitudinal cohorts, we sought to identify the principal predictors of subsequent change across the broader mental health system, focusing on whether screen-related behaviours or psychological difficulties occupied the leading CLPN

net source positions. Within SEARCH, we also examined whether these longitudinal positions corresponded to the parts of the symptom system identified by static structural-control analyses. The pooled GMP analysis provided a cross-sectional comparison of the relative network prominence of emotional and screen-related domains.

Results

Emotional distress is the strongest net source in SEARCH

After data-quality screening, 10,356 adolescents from the School-based Evaluation Advancing Response for Child Health (SEARCH) cohort comprised the Wave 1 analytical sample. Of these, 6,092 were successfully linked to Wave 3, conducted approximately one year after Wave 1, and had complete data across all six longitudinal domains. For consistency across the text and figures, these domains are denoted throughout as CIAS (generalised problematic digital use), DASS (emotional distress), SDQ (broad difficulties), DIET (diet), ISI (insomnia) and ACT (physical activity). Compared with participants who could not be linked, the retained sample had lower baseline scores for CIAS (standardised mean difference, -0.35), DASS (-0.35), ISI (-0.31), DIET (-0.26) and SDQ (-0.18), and slightly higher physical activity (0.10 ; Supplementary Table S12).

Within this retained sample, emotional distress showed the largest positive CLPN net directionality ($+0.249$, 95% CI 0.086 to 0.418 ; Fig. 1), placing it in the strongest net source position. Broad SDQ difficulties (-0.090 , 95% CI -0.189 to -0.003) and physical activity (-0.145 , 95% CI -0.218 to -0.060) occupied net sink positions. By contrast, problematic digital use (CIAS; -0.061 , 95% CI -0.161 to 0.051), diet (-0.014 , 95% CI -0.137 to 0.108) and insomnia ($+0.061$, 95% CI -0.091 to 0.217) showed no significant net directionality because their confidence intervals included zero.

This domain-level pattern was also evident in the individual cross-domain paths. Emotional distress predicted later change in problematic digital use (DASS $\rightarrow$ CIAS, $\beta=0.160$), insomnia (DASS $\rightarrow$ ISI, $\beta=0.151$), broad SDQ difficulties (DASS $\rightarrow$ SDQ, $\beta=0.120$) and diet (DASS $\rightarrow$ DIET, $\beta=0.108$). The similarly strong reciprocal path from insomnia to emotional distress (ISI $\rightarrow$ DASS, $\beta=0.150$) indicated reciprocal prospective prediction between these two domains, while emotional distress retained the strongest positive net directionality. Emotional distress retained the strongest net source position in sensitivity analyses restricted to Bonferroni-significant cross-domain paths, adjusted for age, sex and grade, accounting for clustering within schools, and excluding diet (Supplementary Tables S13). Across these analyses, emotional distress was the leading longitudinal predictor in SEARCH; problematic digital use showed no clear net directionality.

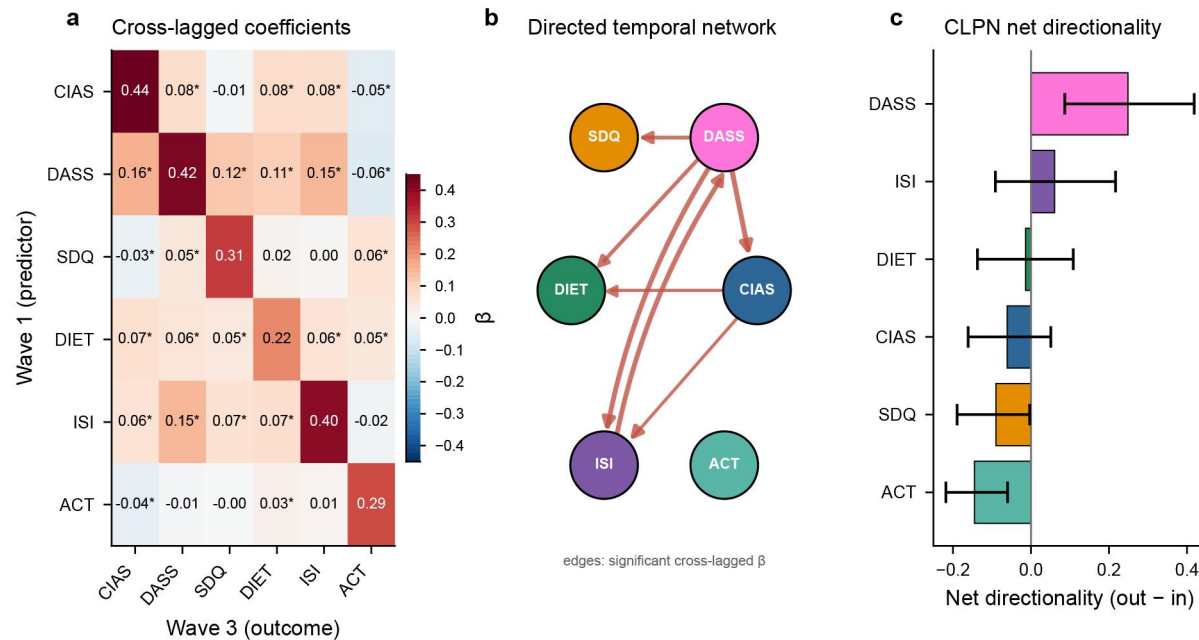


Figure 1 | Longitudinal cross-lagged network in SEARCH (a) Standardised coefficients from each Wave 1 domain to each Wave 3 domain; diagonal entries are autoregressive effects, and asterisks indicate nominal $P < 0.05$ without multiplicity adjustment. (b) Directed network displaying cross-lagged paths with nominal $P < 0.05$ and $|\beta| \geq 0.08$. This display threshold was used for visual clarity. (c) CLPN net directionality, calculated as absolute cross-lagged out-strength minus in-strength, with 95% bootstrap confidence intervals. CIAS, generalised problematic digital use; DASS, emotional distress; SDQ, broad SDQ difficulties; DIET, diet; ISI, insomnia; ACT, physical activity. $N = 6,092$.

Social media use and gaming occupy net sink positions in the UK cohort

In SEARCH, emotional distress occupied the strongest net source position, whereas problematic digital use showed no significant net directionality. We next tested this longitudinal ordering in the independent UK Millennium Cohort Study (MCS), which followed adolescents from age 14 to 17 and directly measured social media use and electronic gaming. Among 9,520 cohort members linked across the two ages, 7,838 had complete data across all six domains and entered the CLPN analysis. The resulting pattern placed screen-related behaviours in net sink positions: social media use was the strongest net sink (-0.211 , 95% CI -0.260 to -0.139), while gaming (-0.089 , 95% CI -0.149 to -0.020) and peer problems (-0.080 , 95% CI -0.161 to -0.004 ; Fig. 2) also occupied net sink positions.

Among the net source positions, hyperactivity/inattention was the strongest ($+0.177$, 95% CI 0.105 to 0.255), followed by self-reported internalising distress ($+0.105$, 95% CI 0.023 to 0.154) and conduct problems ($+0.098$, 95% CI 0.015 to 0.191). The strongest prospective paths to later screen-related behaviour originated from these broader difficulties: hyperactivity/inattention predicted gaming ($\beta = 0.137$) and social media use ($\beta = 0.112$), while conduct problems predicted both social media use ($\beta = 0.080$) and gaming ($\beta = 0.060$). The MCS results broadened the SEARCH pattern: emotional distress was the strongest net source in SEARCH, while MCS identified both internalising and externalising net source

positions, led by hyperactivity/inattention. Across both cohorts, the core ordering was consistent: screen-related behaviours did not occupy leading net source positions, with problematic digital use showing no significant net directionality in SEARCH and social media use and gaming occupying net sink positions in MCS.

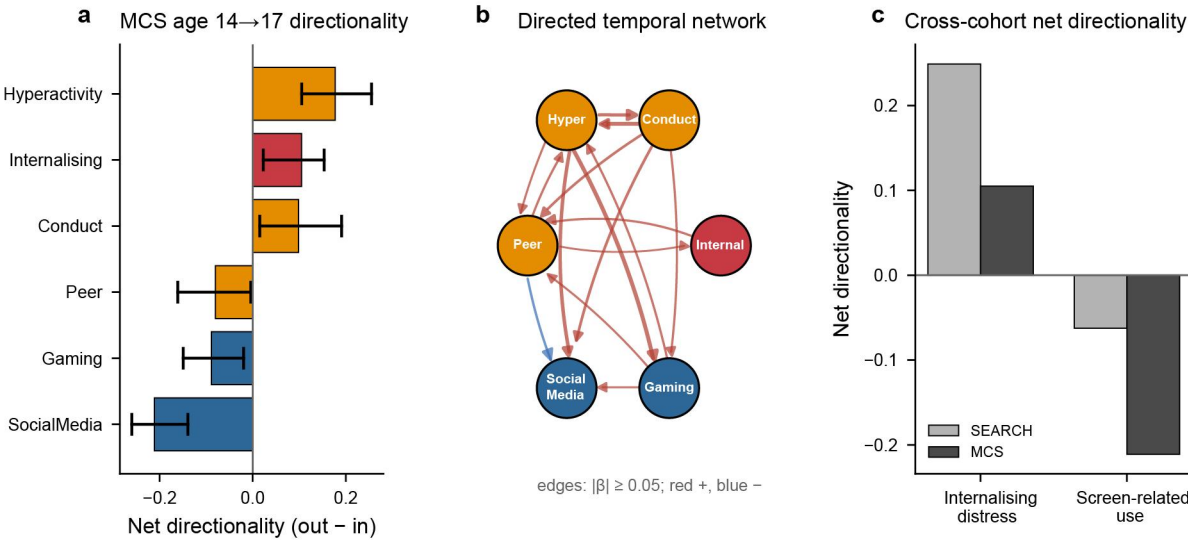


Figure 2 | Longitudinal directionality in the UK Millennium Cohort Study. (a) CLPN net directionality from age 14 to age 17, calculated as absolute cross-lagged out-strength minus in-strength, with 95% bootstrap confidence intervals. Positive values indicate a stronger balance of outgoing over incoming prediction. (b) Directed temporal network showing cross-lagged paths with $|\beta| \geq 0.05$; this display threshold was used for visual clarity and does not denote statistical significance. Red and blue edges indicate positive and negative coefficients, respectively. (c) Cross-cohort comparison of CLPN net directionality for psychological and screen-related domains. SEARCH uses DASS-21 for emotional distress and CIAS-R for generalised problematic digital use; MCS uses self-reported internalising distress and direct measures of social media use and gaming. $N = 9,520$ linked across ages 14 and 17; $n = 7,838$ complete cases included in the cross-lagged analysis.

Static structural control differs from temporal prediction

The longitudinal analyses in SEARCH and MCS placed screen-related behaviours outside the leading CLPN net source positions. We next examined the baseline SEARCH network to identify item-level structural-control positions. The network contained 104 items connected by 2,777 non-zero edges (density = 0.518, modularity = 0.443, global efficiency = 0.759). Despite the dense interconnection, ten equivalent MDS solutions were identified, each comprising three nodes whose immediate neighbourhoods covered the remaining network. Across these solutions, the most recurrent control nodes were DIET22 (worry about eating too much; control frequency = 0.60), SDQ18 (often lies or cheats; 0.50) and DASS14 (intolerance of interruption; 0.40; Fig. 3). The leading control positions arose from diet, behavioural and emotional symptoms. No CIAS item ranked among the three highest-frequency control nodes. These structural-control nodes were not the most highly connected symptoms: control frequency correlated only weakly with node strength (Spearman $\rho = 0.061$) and loaded on a separate component from conventional

centrality measures (Supplementary Fig. S15 and Table S10). Structural coverage and network connectedness identified distinct symptom roles.

Null-model analyses showed that the distinctive finding concerned control-node identity, not MDS size. A three-node minimum set was compatible with the network's degree sequence and was not itself exceptional. DIET22, SDQ18 and DASS14 nevertheless entered the observed solutions more frequently than expected from their node-specific null domination frequencies of 0.186, 0.252 and 0.246, respectively (Supplementary Fig. S19). The exact solution remained sensitive to network estimation. Sparser EBIC graphical lasso networks produced five- or six-node MDS solutions and altered membership rankings. The primary cross-validated network retained the three-node solution (Supplementary Table S17 and Fig. S20).

At the broader domain level, diet and broad SDQ difficulties occupied the strongest structural-control positions, emotional distress and problematic digital use formed an intermediate layer, and physical activity and insomnia occupied the weakest positions. This static architecture did not reproduce the longitudinal hierarchy. Emotional distress held an intermediate structural position but occupied the strongest CLPN net source position. Problematic digital use remained intermediate in the static network and showed no significant CLPN net directionality. Static structural leverage and CLPN net directionality captured different dimensions of the symptom system.

The broader six-domain architecture remained highly stable across complementary robustness checks. Six community-detection algorithms recovered the same domain organisation (adjusted Rand index, 0.61 to 0.73; normalised mutual information, 0.78 to 0.84), with the Leiden algorithm returning exactly six communities³¹. Among the 89 items that could be compared across waves, community agreement remained high (adjusted Rand index = 0.86), with only eight items changing community. Bridge centrality and structural-control frequency also identified different nodes³², reinforcing the distinction between symptoms that connect domains and those that govern coverage of the wider network. Together, these findings support the stability of the six-domain architecture and its distinct structural-control dimension, while localising estimator sensitivity to the membership of individual MDS solutions.

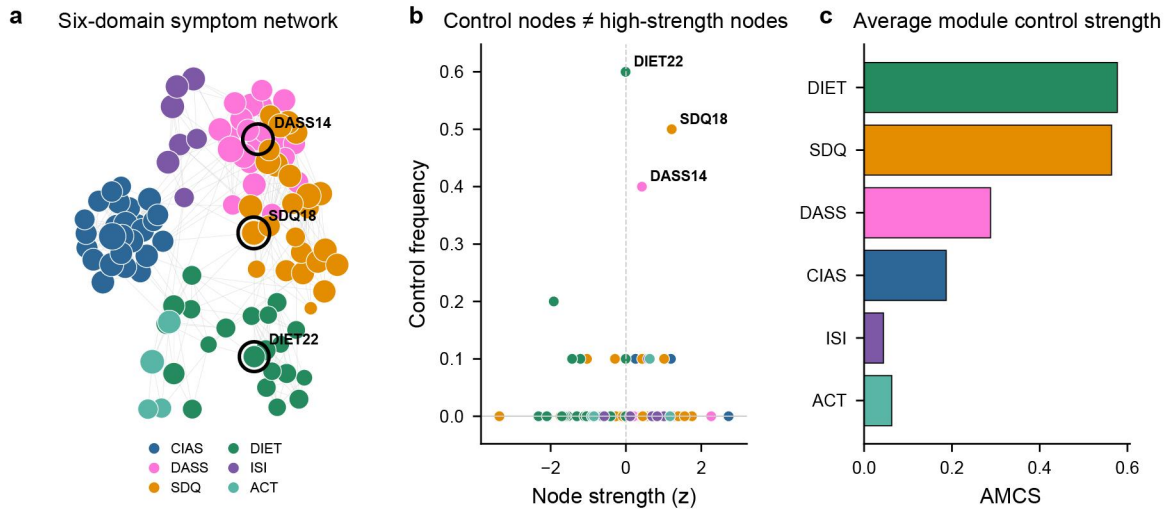


Figure 3 | Baseline structural network and structural-control profile in SEARCH. (a) Gaussian graphical model showing the 500 largest edges; node colour denotes domain, node size denotes strength and rings identify the three highest control-frequency nodes. (b) Node strength versus control frequency. (c) Average module control strength (AMCS) across the six domains. Structural-control quantities describe the topology of the baseline undirected graph. $N = 10,356$. All analyses used the full estimated network.

Bayesian and CLPN rankings partially align

An exploratory Bayesian directed acyclic graph fitted to the six Wave 1 SEARCH domain scores was used to compare Bayesian net out-degree with CLPN ordering. DASS had the highest net out-degree (+2), with outgoing edges to SDQ, CIAS and DIET. By contrast, CIAS had balanced in- and out-degree, while DIET had the most negative Bayesian net out-degree (-2 ; Fig. 4). The most stable directed edges were $\text{CIAS} \rightarrow \text{ACT}$ (bootstrap frequency = 0.970), $\text{DASS} \rightarrow \text{SDQ}$ (0.910), $\text{CIAS} \rightarrow \text{DIET}$ (0.905), $\text{DASS} \rightarrow \text{DIET}$ (0.855) and $\text{DASS} \rightarrow \text{CIAS}$ (0.850). Three of these five leading edges originated from DASS.

Across the six domains, Bayesian net out-degree was positively correlated with CLPN net directionality, although the estimate was imprecise (Pearson $r = 0.750$, two-sided $P = 0.086$; bootstrap 95% CI -0.604 to 0.995). DASS had a positive Bayesian net out-degree and occupied the strongest CLPN net source position. All seven skeleton edges were reversible in the CPDAG (Supplementary Tables S15–S16). These results show partial correspondence in domain ordering between the Bayesian and CLPN models.

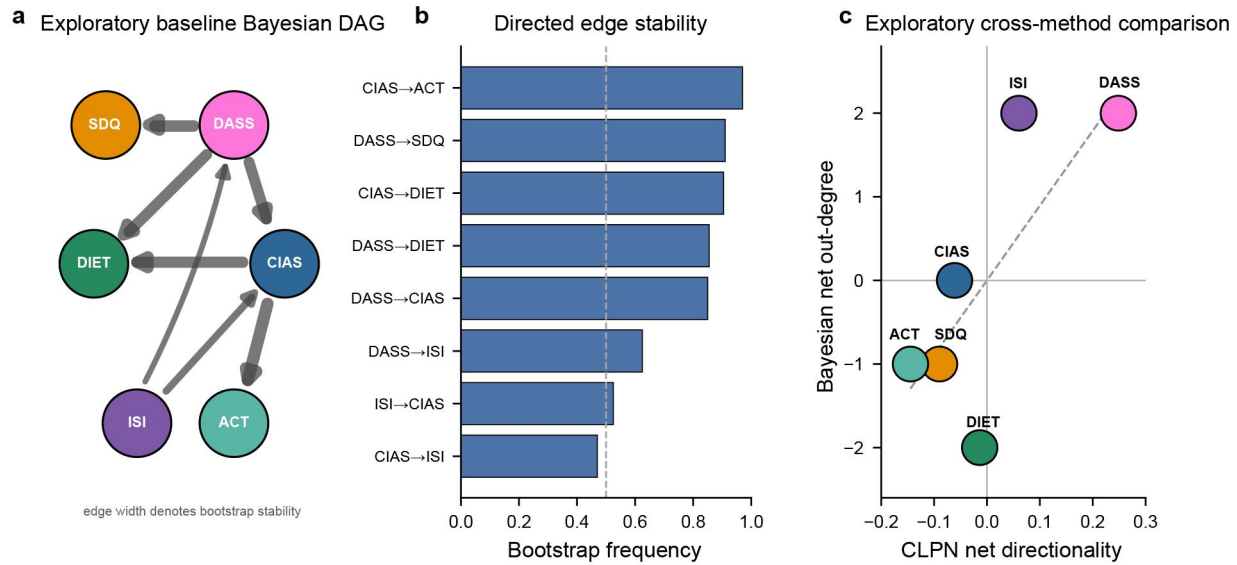


Figure 4 | Exploratory Bayesian net out-degree and CLPN net directionality in SEARCH. (a) Exploratory Bayesian directed acyclic graph learned from tertile-discretised baseline domain scores; edge width denotes bootstrap stability. (b) Bootstrap frequencies of the eight most stable directed edges. (c) Association between Bayesian net out-degree and longitudinal cross-lagged panel network net directionality across the six SEARCH domains. CIAS, generalised problematic digital use; DASS, emotional distress; SDQ, broad difficulties; DIET, diet; ISI, insomnia; ACT, physical activity; DAG, directed acyclic graph; CLPN, cross-lagged panel network. N = 10,356 for the baseline DAG; n = 6,092 for the longitudinal CLPN; panel c contains six paired domain-level estimates.

Emotion ranks highest in network strength in GMP

We examined whether the emotional prominence identified in SEARCH held under a different measurement system and across a much broader population. The Global Mind Project (GMP) analysis included 147,506 participants aged 18–24 years with complete data across six mapped domains from a pooled multinational sample. Emotion had the highest undirected network strength (0.726, 95% CI 0.714 to 0.738). It was followed by diet (0.504, 95% CI 0.493 to 0.515), activity (0.444, 95% CI 0.434 to 0.455), broad behavioural difficulties (0.409, 95% CI 0.396 to 0.424) and sleep (0.409, 95% CI 0.401 to 0.420). At the opposite end of the network, the social-media-frequency proxy had the lowest strength (0.075, 95% CI 0.065 to 0.085; Fig. 5), placing social-media frequency at the periphery of the mapped health system.

For the construct-level comparison, a corresponding six-domain SEARCH network was derived from the baseline domain scores. After conventional undirected strength was normalised within each of these separate six-domain comparison networks, emotion ranked first in both SEARCH and GMP. In SEARCH, emotion was followed by generalised problematic digital use, insomnia, broad SDQ difficulties, diet and physical activity. Social media frequency was the weakest domain in GMP (Fig. 5b). This descriptive strength comparison was distinct from the item-level structural-control analysis reported above.

Across these distinct populations and analytical frameworks, emotional difficulties remained prominent, while screen-related measures occupied less prominent positions.

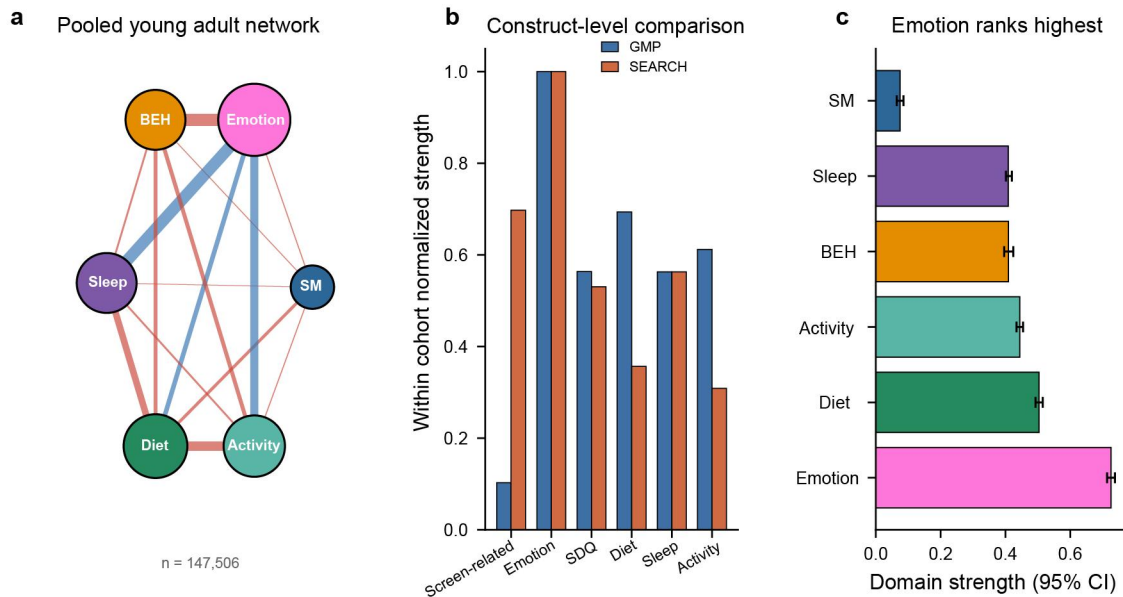


Figure 5 | Cross-sectional construct-level comparison in GMP. (a) Six-domain Gaussian graphical model in young adults; node size denotes domain strength. (b) Within-cohort normalised domain strength in the six-domain GMP network and a corresponding six-domain network derived from aggregated baseline SEARCH scores. Domain strengths were divided by the maximum strength within each cohort, so that the strongest domain was assigned a value of 1. (c) GMP domain strength with 95% bootstrap confidence intervals. In GMP, the screen-related domain was represented by social media frequency. Emotion had the highest network strength in GMP. N = 147,506 participants aged 18–24 years in the pooled multinational GMP sample. BEH, broad behavioural difficulties; SM, social-media frequency.

Discussion

Across both cohorts, broader emotional and behavioural difficulties were the leading longitudinal predictors; screen-related measures did not occupy leading net source positions. In SEARCH, emotional distress had the largest positive CLPN net directionality and predicted later change in generalised problematic digital use, insomnia, broad SDQ difficulties and diet; problematic digital use itself had no significant net directionality. In MCS, directly measured social media use was the strongest net sink and gaming also occupied a net sink position. The net source positions included hyperactivity/inattention, internalising distress and conduct problems, extending beyond emotional distress alone. Across cohorts, broader psychological difficulties occupied the leading net source positions, although the specific domains differed. Problematic digital use, social media use and gaming did not occupy leading CLPN net source positions when analysed within broader psychological systems.

This distinction changes how associations between digital use and mental health should be interpreted. Elevated problematic digital-use scores mark clinically relevant functional impairment and warrant assessment in their own right. Their network position also places digital use within the accompanying symptom burden, outside the leading predictive position. In SEARCH, emotional distress occupied the strongest net source position; in MCS, externalising difficulties, especially hyperactivity/inattention,

predicted later screen-related behaviour. Clinical and preventive assessment should therefore prioritise the emotional, attentional and conduct context of escalating digital engagement rather than assessing screen exposure in isolation. These estimates define prospective relations that can be tested in intervention and intensive longitudinal designs^{25, 33-35}.

The comparison of static and temporal models provides a second contribution. MDS analysis identified diet, SDQ and emotional items whose neighbourhoods covered the baseline graph; the CLPN placed DASS in the strongest domain-level net source position. Control frequency was only weakly related to node strength, and average module control strength did not reproduce CLPN net directionality. Graph topology and longitudinal prediction identified different dimensions of the symptom system. Structural-control nodes should consequently be interpreted as features of an estimated cross-sectional graph, not as causal intervention targets. Their value lies in describing sparse coverage and architecture; temporal or causal prioritisation requires longitudinal or experimental evidence²⁰⁻²⁵. In the exploratory Bayesian graph, DASS had the highest net out-degree, which was positively but imprecisely associated with CLPN net directionality across domains.

The external analyses sharpen the construct boundary. SEARCH assessed generalised problematic internet and digital-media use through the CIAS-R, whereas MCS measured hours of social media use and gaming, and the Global Mind Project used social-media frequency. These measures provide construct-level convergence across related forms of digital use without implying instrument-level replication. The MCS variables provided the longitudinal exposure-based comparison: both social media use and gaming occupied net sink positions. The Global Mind Project provided a pooled multinational cross-sectional comparison: emotion had the highest network strength, whereas social-media frequency had the lowest. These findings distinguish impairment-oriented problematic use from exposure-oriented frequency: screen-related measures did not occupy leading CLPN net source positions and had relatively low cross-sectional network strength in GMP.

Several properties of the data define the scope of inference. Because the SEARCH model used two waves and estimated between-person lagged prediction, it cannot separate stable person differences from within-person dynamics^{36, 37}. Linkage retained 6,092 of 10,356 baseline participants, and the retained sample had lower baseline symptom burden, which may attenuate or alter longitudinal estimates. The broad SDQ domain included emotional, conduct, hyperactivity, peer and reverse-scored prosocial items; DASS → SDQ paths therefore represent prediction of broad difficulties rather than conduct-specific change and partly span overlapping affective content. Diet response options changed between waves; wave-standardised domain scores and a five-domain sensitivity analysis addressed this change. MCS used different internalising instruments at ages 14 and 17, precluding scale-level invariance testing. Finally, the

pooled Global Mind Project analysis does not estimate country-specific networks and remains cross-sectional.

The evidence supports a predictive reframing of adolescent digital-use research. Screen-related measures remained outside the leading CLPN net source positions: problematic digital use showed no significant net directionality in SEARCH, while social media use and gaming occupied net sink positions in MCS. Emotional distress occupied the strongest net source position in SEARCH, while internalising and externalising difficulties occupied net source positions in MCS. Static network control identified a different set of topological features. Together, these findings position screen-related behaviours within the broader psychological context of adolescent mental health and support integrated assessment of digital engagement alongside emotional and behavioural difficulties.

Methods

Participants and study design

Data came from the School-based Evaluation Advancing Response for Child Health (SEARCH), a mixed longitudinal school-based cohort conducted across 11 schools in three cities of Jiangsu Province, China²⁷. The baseline wave (Wave 1) was collected from September 2022 to February 2023. Of 11,640 unique records in the analytic export, 1,284 (11.0%) were excluded because either the participant-level median item-response time or the within-participant standard deviation across the 104 analysed item scores fell outside sample-derived bounds of $1.5 \times$ the interquartile range. The baseline sample comprised 10,356 adolescents (52.2% male; mean age 13.96 years, SD 2.42). The follow-up wave used in the longitudinal analysis (Wave 3) was exported in April 2024, approximately one year after baseline. The study was approved by the Ethics Committee of the Affiliated Brain Hospital of Nanjing Medical University (approval 2022-KY095-02), was conducted in accordance with the Declaration of Helsinki, and obtained written informed consent from participating adolescents and their legal guardians.

SEARCH measures

We assessed six domains with 113 items. Problematic digital use was measured with the 26-item Chen Internet Addiction Scale-Revised (CIAS-R; response range 1–4), which assesses impairment across internet and digital-media use rather than smartphone-specific behaviour³⁸. Emotional distress was measured with the 21-item Depression Anxiety Stress Scales (DASS-21; range 0–3)³⁹. Broad difficulties were assessed with the 25 core SDQ items; six impact-supplement items were recorded separately⁴⁰. Insomnia was measured with the seven-item Insomnia Severity Index (ISI; range 0–4)⁴¹. Diet was assessed using nine cohort-specific dietary-behaviour items and 13 items from the Chinese modified Yale Food Addiction Scale 2.0^{42, 43}. Physical activity was assessed with six items; items 1–4 were adapted from

the International Physical Activity Questionnaire-Short Form^{44, 45}, and items 5–6 assessed recreational screen time. The exclusions used to construct the 104-item network are described below.

Nine items were excluded before item-level network estimation. The six SDQ impact-supplement items (items 26–31) were excluded because they assess the duration, distress and functional impact of overall difficulties and do not represent individual symptoms. DIET item 3 was a multiple-response item and was not comparable with scalar items. ACT items 5 and 6 assessed recreational screen time and were excluded from the physical-activity domain. After these exclusions, the final baseline network contained 104 items: 26 CIAS-R, 21 DASS-21, 25 SDQ, 7 ISI, 21 DIET and 4 ACT items. SDQ items 7, 11, 14, 21 and 25 were reverse-scored so that higher values indicated greater difficulty. Because all 25 core SDQ items were retained, including emotional and reverse-scored prosocial content, we describe this domain as broad SDQ difficulties. The CIAS-R, DASS-21, SDQ and ISI have established psychometric properties.

SEARCH longitudinal linkage and domain scores

The two SEARCH waves used independent record identifiers. Participants were linked deterministically using a SHA-1 digest of name, date of birth and recorded sex, truncated to 16 hexadecimal characters. Composite keys that were non-unique within a wave were excluded before linkage; three within-wave duplicate records in baseline and one at follow-up were removed. This procedure produced 6,092 participants with complete six-domain data at both waves. Non-unique baseline linkage keys involved 0.052% of candidate records.

Diet items 10-22 changed from binary response options at baseline to an eight-point scale at follow-up. Accordingly, the primary longitudinal analysis used six domain scores; item-level diet comparisons were not performed. Each domain score was z-standardised within wave. Cross-wave correlations were 0.58 for CIAS-R, 0.63 for DASS, 0.43 for SDQ, 0.34 for diet, 0.57 for ISI and 0.31 for physical activity. Baseline differences between linked and unlinked participants were quantified with standardised mean differences.

SEARCH cross-lagged panel network

For each follow-up domain, we fitted an ordinary least-squares regression containing all six baseline domain scores, including the corresponding autoregressive predictor. Coefficients were standardised and HC3 heteroskedasticity-robust standard errors were used^{26, 36}. The 30 cross-domain paths were evaluated using a Bonferroni-adjusted threshold of $\alpha = 0.0017$. For visualisation, Fig. 1a marks paths with nominal $P < 0.05$, and Fig. 1b displays paths with nominal $P < 0.05$ and $|\beta| \geq 0.08$. Formal inference was based on the Bonferroni-adjusted threshold. CLPN net directionality was calculated for each domain as the sum of absolute cross-lagged outgoing coefficients minus the sum of absolute incoming coefficients; autoregressive coefficients were excluded. Positive values indicate a net source position and negative values a net sink position. Because the calculation uses absolute coefficients, both positive and negative

cross-lagged paths contribute to net directionality. Percentile 95% confidence intervals were estimated with 1,000 nonparametric bootstrap resamples.

Sensitivity analyses recalculated directionality using only Bonferroni-significant paths, adjusted the regressions for age, sex and grade, used school-cluster-robust inference, and fitted a five-domain model excluding diet. These analyses tested the robustness of the net source–sink ranking and directionality sign under alternative specifications.

Millennium Cohort Study longitudinal analysis

The UK Millennium Cohort Study (MCS) was used for an external longitudinal analysis²⁸. Sweep 6 at age 14 (N = 11,852)⁴⁶ and Sweep 7 at age 17 (N = 10,343)⁴⁷ were obtained under licence from the UK Data Service. De-identified family and cohort-member identifiers linked 9,520 adolescents across sweeps. Six nodes were available at both ages: self-reported internalising distress, assessed with the 13-item Short Mood and Feelings Questionnaire at age 14⁴⁸ and the Kessler K6 at age 17⁴⁹; SDQ conduct, hyperactivity/inattention and peer-problem subscales; self-reported hours of social media use; and self-reported hours of electronic gaming. Scores were z-standardised within wave.

The MCS CLPN used the six constructs available at both sweeps and followed the same regression specification and net-directionality definition as SEARCH, with 500 nonparametric bootstrap resamples. The complete-case analysis included 7,838 participants. Because internalising distress was measured using different instruments at ages 14 and 17, this node was interpreted at the construct level.

Baseline SEARCH network reconstruction

Item responses were transformed to normal marginals with a nonparanormal Gaussian-copula transformation. A Gaussian graphical model was estimated with graphical lasso^{18, 19}. The regularisation parameter was selected by likelihood-based K-fold cross-validation over a 100-point geometric grid from $\lambda_{\max} = 0.824$ to 1% of that value, yielding $\lambda = 0.0581$. Partial correlations were derived from the precision matrix. Cross-sectional network reporting followed current methodological standards²³.

Structural-control and robustness analyses

MDS solutions were enumerated exhaustively^{20, 21}. A dominating set is a group of nodes such that every node in the graph either belongs to the set or is adjacent to a member. Control frequency was defined as the proportion of MDS solutions containing a node. The module control network summarised inter-module structural coverage using average module control strength (AMCS)²². These quantities refer to the topology of the estimated undirected graph; they do not represent intervention effects, dynamical controllability or temporal precedence.

To separate minimum-set size from node identity, 500 degree-preserving null networks were generated by double-edge-swap randomisation and each node's observed control frequency was compared with its node-specific null domination frequency. Estimator sensitivity was examined with EBIC graphical lasso networks at gamma values of 0, 0.25 and 0.5, the primary cross-validated network, and a positive-edges-only version. Six community-detection algorithms (Louvain, Leiden, Walktrap, Spinglass, Fast-Greedy and Infomap) were compared with the six prespecified domains using adjusted Rand index and normalised mutual information³¹. Community assignments were also compared across waves.

We additionally calculated node strength, closeness, betweenness, eigenvector centrality, signed expected influence, bridge strength, bridge expected influence, bridge betweenness, participation coefficient, average controllability, modal controllability and control energy³². Average and modal controllability, together with the control energy associated with state transitions in the ISI and ACT modules, were derived from the discrete-time linear controllability Gramian applied to the absolute-weight adjacency matrix. Spearman correlations and principal-component analysis assessed overlap among node metrics. Node-knockout analyses and sustained nonlinear propagation simulations were retained as supplementary robustness analyses. The propagation model applied a hyperbolic-tangent activation with gain $g = 0.9$ to the row-normalised absolute adjacency matrix under sustained module input. Group-level network structures were compared using a permutation-based network comparison test⁵⁰.

Exploratory Bayesian graph

The six baseline SEARCH domain scores were discretised into tertiles and a directed acyclic graph was learned by hill-climb search with the Bayesian information criterion in pgmpy. Directed-edge stability was estimated across 200 bootstrap resamples. We also examined the completed partially directed acyclic graph, quartile discretisation, a continuous Gaussian Bayesian network and leave-one-domain-out correlations. Because observational Bayesian graphs identify an equivalence class and may be affected by unmeasured confounding, this analysis was interpreted as an exploratory comparison with CLPN ordering rather than causal identification.

Global Mind Project analysis

The Global Mind Project/Mental Health Million dataset contains adult self-report records collected internationally^{29, 30}. We selected participants aged 18–24 years; 147,506 had complete data for six mapped constructs and were included in the pooled multinational analysis. Emotion was represented by fear/anxiety, sadness/hopelessness, mood swings and anger; broad behaviour by restlessness/hyperactivity, aggression and impulsivity; sleep by sleep frequency and quality; diet by ultra-processed food, fruit/vegetable and sugary-food frequency and appetite; activity by exercise frequency; and digital exposure by social-media frequency.

Within each mapped GMP domain, component items were z-standardised and averaged to form a domain score. The six domain scores were transformed to normal marginals, and a Gaussian graphical model was estimated using cross-validated graphical lasso. Domain strength was calculated as the sum of absolute partial correlations, with 95% confidence intervals obtained from 300 nonparametric bootstrap resamples.

For the descriptive comparison in Fig. 5b, the 104 baseline SEARCH items were summed within six corresponding domains, and the resulting domain scores were z-standardised. A six-domain partial-correlation network was derived from the inverse correlation matrix, with domain strength calculated in the same way. Strengths were normalised separately within each cohort by dividing each value by the maximum domain strength, so that the strongest domain was assigned a value of 1. This comparison was descriptive and did not test differences between cohorts.

Statistics and reproducibility

All available eligible records were analysed and no a priori sample-size calculation was performed. No missing-data imputation was used. The SEARCH cross-lagged model included 6,092 linked complete cases, the MCS model included 7,838 complete cases, the secondary baseline SEARCH network included 10,356 participants, and the Global Mind Project analysis included 147,506 complete cases. All tests were two-sided. Bootstrap counts were 1,000 for SEARCH directionality, 500 for MCS directionality, 300 for Global Mind Project strength, 200 for Bayesian edge stability and 500 for degree-preserving network nulls.

Software

Analyses were conducted in Python using scikit-learn, networkx, python-igraph, leidenalg, python-louvain, statsmodels and pgmpy.

Data Availability

Individual-level SEARCH data are restricted by ethics approval 2022-KY095-02 and cannot be shared publicly. Global Mind Project data are available from Sapien Labs under the provider's data-access conditions. Millennium Cohort Study data are available to registered users from the UK Data Service under its End User Licence. Derived, non-identifiable matrices, cross-lagged coefficients, bootstrap outputs, Bayesian-network edge lists, null-model distributions and machine-readable result tables are supplied in Supplementary Data 1 and the accompanying code archives. No individual-level records are included.

Code Availability

Analysis scripts for SEARCH network reconstruction, cross-lagged panel networks, structural-control analyses, Bayesian graphs, community detection, null models and figure generation are supplied in the

accompanying main code archive. External Global Mind Project and Millennium Cohort Study scripts are supplied in a separate external-validation code archive.

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Author Contributions

X.Z. conceived and designed the study. J.Z. and L.Z. curated the data. X.Z., J.Z. and L.Z. conducted the statistical and network analyses. X.Z. supervised the work. J.Z., L.Z. and X.Z. interpreted the findings. X.Z. drafted the manuscript. All authors reviewed and approved the final manuscript.

Ethics Declaration

The present study analysed existing SEARCH, MCS and GMP data; no new participants were recruited and no new data were collected. The SEARCH study was approved by the Ethics Committee of the Affiliated Brain Hospital of Nanjing Medical University (approval 2022-KY095-02) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from participating adolescents and their legal guardians before data collection. The Global Mind Project and Millennium Cohort Study analyses used de-identified secondary data collected under the ethical approvals and access conditions of their respective data providers.

Competing Interests

The authors declare no financial or non-financial competing interests.

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